A local moment refinement near the quartic zero-free boundary

Zhongpai Gao*

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Abstract

We give a proposed refinement of the moment estimate in Collet's quartic zero-free argument. Near the lengths used by the weighted numerator, the coefficient of the prime-slot length is reduced from 9/2 to 4.498. The proof separates rows according to their zero parameter: lower rows are paid by removing slots, whereas higher rows are counted jointly with their arithmetic labels. An exact rational certificate then gives the boundary 0.8748866 for the Eisenstein Hecke and ordinary Dirichlet families. This paper presents the argument and its quantitative margins; the accompanying supplement retains the full source-level proof, including masks, transformed children, and the continuation argument.

1 The result and the moment family

The starting point is the pinned Cycle25 argument of Collet [1], built on OpenAI's 7/8 theorem [2] and Levy's weighted numerator [3]. Its comparison boundary is

$$b_0 = \frac{11}{12} - \frac{\ell_0}{4} = 0.8748866470399641\ldots, \qquad 927\ell_0^4 - 3135\ell_0^3 + 2433\ell_0^2 + 275\ell_0 - 100 = 0,$$

where $\ell_0 \in [1/6, 1/5]$. We modify one local moment estimate in that argument and leave the detector and weighted Euler expansion unchanged.

Theorem 1.1 (Proposed endpoint, relative to the pinned framework)**.** *Put $\theta = 4374433/5000000 = 0.8748866$. Every finite-order Hecke L-function of trivial infinite type over $\mathbb{Q}(\sqrt{-3})$ is zero-free in $\Re s > \theta$, with the principal pole at $s = 1$ excluded. The same conclusion holds for every ordinary Dirichlet L-function, including the Riemann zeta function. Characters may be imprimitive and have arbitrary conductor.*

The local input is easiest to state in the Eisenstein ring $\mathcal{O} = \mathbb{Z}[e^{2\pi i/3}]$. Let $U = Z^M$, $Z > 1$, $M > 0$, and let N denote the norm. In the natural family of the source, a row k acts by

$$\psi_k(n) = \tau(n)\chi_n(k),$$

where τ is a fixed finite ray character and $\chi_n(k)$ is the sextic residue character with its natural zero extension. There is no parent character of positive moving width. Write $S_{k,j}(Z^{n_j})$ for the centrally

*Research, computation, formalization, and exposition received AI assistance. Mathematical credits and pinned sources are given in Section 6.

normalized smooth ordinary character sum on scale Z^{n_j}. A prime slot is

$$Q_{k,i} = Z^{-z_i/2} \sum_{\mathfrak{p} \in \mathcal{P}_i^{\mathrm{can}}} \psi_k(\mathfrak{p}) \nu_i(\mathfrak{p}) W\left(\frac{N\mathfrak{p}}{Z^{z_i}}\right) \left(\frac{N\mathfrak{p}}{Z^{z_i}}\right)^{-\sigma_i + iv_i}. \tag{1}$$

The pools $\mathcal{P}_i^{\mathrm{can}}$ are the source's canonical good-prime pools. The slots share one smooth annular profile W, while their allowed relative characters ν_i and real powers may differ, and their heights are independent. Set $z = \sum_i z_i$ and $Q_{k,I} = \prod_i Q_{k,i}$.

All natural zeros are retained. We also allow a common nonzero puncture of norm at most U^B, independent of the summed row but possibly dependent on frozen outer labels. A row is *eligible* precisely when it is not a fixed-inducing presentation in the source sense. The exclusion ideal

$$Q_* = M_{\mathrm{ray}} \cap (72) \cap \mathfrak{q}_{\eta_0}$$

is chosen with the fixed family and anchor η_0; it is proper, nonzero, and stays fixed under transformations. This excludes all principal twists within the actual parent translate of the allowed Q_*-character group. It does not identify different parent translates. Whole-character conjugation is handled by height reflection (Supplement C).

Proposition 1.2 (Local physical moment)**.** *Assume the incumbent bound $\beta_* \le 7/8$ and its nontrivial branch $\beta_* \ge 51/100$. In the preceding fixed family, put $C = 2249/500$. If*

$$.41M \le n_i \le .44M, \qquad .03M \le z \le .04M, \qquad n_1 + n_2 + Cz \le M, \tag{2}$$

then, for every $\varepsilon > 0$, sufficiently fine canonical slots satisfy

$$\sum_{k \text{ eligible}} |S_{k,1}(Z^{n_1}) S_{k,2}(Z^{n_2}) Q_{k,I}|^2 \ll Z^{M+\varepsilon} \mathcal{H}_J. \tag{3}$$

The source radial weight is understood. Here $\mathcal{H}_J$ consists of finite smooth seminorms and polynomial factors in all independent heights. The estimate includes the four derivative-profile pairs used for two-variable scale-supremum recovery.

The slot mesh is chosen from the requested loss and bounded geometric ranges before the live arithmetic labels. For each fixed family and slot system, profile orders, height orders, constants, and the eventual threshold precede the live scales, masks, and heights. The descendants are estimated by the existing 9/2 and unrestricted zero-slot moments, not by another application of Proposition 1.2. Sections 2–3 give its proof, with the complete assembly in Supplement G.

2 Rows with a smaller zero parameter

Put

$$A = n_1 + n_2 + z, \qquad g = (n_1 + n_2 + \tfrac{9}{2}z - M)_+, \qquad a = M/2000, \qquad \sigma = 3a.$$

The change of coefficient is only $9/2 - C = 1/500$, so

$$g \le M/12500, \qquad .85M \le A \le 44753M/50000 < .9M. \tag{4}$$

The auxiliary amplifier has prime norms on scale Z^a; its cost must therefore be paid by a fixed positive margin, not absorbed as an arbitrarily small loss.

After the two source transformations, classify nonexceptional rows by a buffered zero parameter δ, whose real zero location is $(1+\delta)/2$. Assign every bin straddling $7/10$ to the lower branch. Its upper endpoint is then $7/10$ plus a small buffer. On this branch, a removed slot of width v costs $(7/10)v$ in the squared-norm exponent, before the reserved buffer losses. It releases $(9/2)v$ of the old moment's capacity.

For a canonical child of actual width M_{act} and clipped plain lengths x_1, x_2, the capacity defect is

$$F_{\mathrm{act}} = (x_1 + x_2 + \tfrac{9}{2}\sum_i z_i - M_{\mathrm{act}})_+.$$

Greedily remove slots of total width at most $F_{\mathrm{act}}/(9/2)$ plus one mesh interval. This either restores the old capacity or removes all slots. The latter case uses the unrestricted zero-slot theorem, which has no residual condition on $x_1 + x_2$.

This payment is made *before* removing the lower-row indicator. The natural coefficient factorization and the nonnegative radial weight give a positive sum of the remaining canonical products. Positivity then enlarges it to the full old moment. Thus neither a discontinuous zero-bin indicator nor an artificially altered coefficient is subjected to a new Poisson transformation (Supplement F).

The dyadic child width need not equal its nominal ledger width. If $d = M_{\mathrm{act}} - M_{\mathrm{nom}} \le \eta_2$ and the nominal defect is at most $F \ge 0$, then $F_{\mathrm{act}} \le (F-d)_+$. For $0 \le r \le 1$,

$$d + r(F-d)_+ \le \eta_2 + rF. \tag{5}$$

Consequently a narrower actual state compensates for its larger defect. Taking $r = (7/10)/(9/2) = 7/45$, the transformed main child has nominal defect at most $g + 6a + \eta$ and source saving at least a. Before the small losses, its reserve is

$$a - \frac{7}{45}(6a+g) \ge \frac{M}{2000} - \frac{7}{45}\left(\frac{3M}{1000} + \frac{M}{10000}\right) = \frac{4M}{225000} > 0. \tag{6}$$

Choose the buffer, mesh overshoot, and support padding within this reserve. Mask invariance of the zero maximum ensures that the pointwise slot estimate applies to each actual allocated child.

3 Higher rows: count the row and its labels together

For $7/10 \le \delta \le 3/4$, the fixed-family unmarked density exponent is

$$\rho(\delta) = 1 - \frac{10\delta}{3(5-2\delta)}, \qquad \frac{2}{7} \le \rho(\delta) \le \frac{19}{54}. \tag{7}$$

Here "unmarked" means that no positive prime-slot supply is required. Supplement D supplies the plain and inverse inputs for the fixed character dictionary; Supplement C identifies its natural representative, conductor, and literal exclusion. The height-uniform density estimate holds on all norm windows, including bounded windows.

To pass from sixth-power-free cores to all rows, write $w = ub^6$. The factors introduced by b are Euler holes: in $\Re s > 0$ they do not change zeros, since $1 - \chi(\mathfrak{p})(N\mathfrak{p})^{-s} \ne 0$. At a common buffered height, retaining its polynomial height factor, a core count $O(Y^{\rho+\varepsilon})$ gives

$$\#\{w : Nw \le H\} \ll H^{\rho+\varepsilon} \sum_b (Nb)^{-6(\rho+\varepsilon)} \ll H^{\rho+\varepsilon}, \qquad H \ge 1.$$

One convergent ideal series with exponent $12/7$ dominates uniformly.

The essential point is to apply this count before freezing the first arithmetic labels. To see which labels the row determines, write $c_n(\pi)$ for the zero-extended sextic row and $q(\pi, n)$ for the fixed reciprocity phase. For $1 \le t \le 5$,

$$c_n(\pi)c_\pi(n)^t = q(\pi,n)^t c_n(\pi)^{1+t},$$
$$c_n(\pi)\overline{c_\pi(n)^t} = q(\pi,n)^{6-t} c_n(\pi)^{7-t}.$$

The two sides therefore have residues $1+t$ and $1-t$ modulo six. When the exponent becomes six, the factor is the mask $\mathbf{1}_{(n,\pi)=1}$, not one. Let F be the product of the chosen primary prime generators to these residues. In the second extraction, G is the common gcd generator and V the forced nonunit radical. The genuine frequency is GVh', and the complete integer row is

$$w = u_0 eFGVh', \qquad \log_Z Nw \le M - R - (c-d)_+ + L - 6a + \eta, \tag{8}$$

with a retained unit u_0 and $L = \log_Z NF$. Here c, d are first common-part widths, R is active conductor width, and e is the extracted mask divisor. The exact identities are proved in Supplement B.

There are two different projectors. The first divides the second common gcd G and hence w. The second need not divide w: it stays fixed outside the row sum, with its full signed allocation and paired normalization. If allocation writes $n = dv$, the exact coefficient is $\psi_w(d)\psi_w(v)$, even when the outside factor vanishes; the quotient receives no new coprimality restriction.

Let p be the first radical width, $E = \log_Z Ne$, and R_0 the width of active primes whose row phase cancels. The first radical absent from w has width at most $p - R - E + R_0$. Once it and w are fixed, all other first and second labels are supported on their product. Rankin's inequality gives the uniform reconstruction estimate

$$\#\{I : NI \le Z^T,\ \operatorname{rad} I \mid Q\} \ll_{B,T,\varepsilon} Z^{\varepsilon} \qquad (NQ \le Z^B).$$

Indeed, bound the count by $Z^{sT}\prod_{\mathfrak{p}\mid Q}(1-(N\mathfrak{p})^{-s})^{-1}$, choose sT small, and absorb the large primes into an arbitrarily small power of NQ. The finitely many small primes contribute a constant. Only the missing radical costs its full norm exponent.

The actual smooth row ball has exponent

$$K_0 + (d-c)_+ + 6a + \varpi, \qquad K_0 = 2A - c - d + R + E - M.$$

The positive padding ϖ includes the first dyadic and fixed support allowances. It contains every amplified row and shortens the dual range in (8). If a fixed parent modulus has width q, write $M = m+q$ and use m in K_0; the final total-width ledger is unchanged. A common Fourier majorant is chosen before the later labels and bins, so the joint count preserves independent heights and the original smooth seminorms (Supplement E).

The resulting excess over the source joint allowance has the form

$$\mathcal{E} = M(\rho - 1) + \delta A + a(5 - 6\rho) + \mathcal{E}_1 + \mathcal{E}_2 + \eta, \tag{9}$$

where the explicit common-part ledger gives

$$\mathcal{E}_1 \le -\frac{79}{270}p - E - B_c, \qquad \mathcal{E}_2 \le -\delta b_2 - t_2, \qquad B_c = (3c - 5d - R)_+/6.$$

These inequalities use $L \le 5(R - R_0)$ and retain every canceled prime in the missing-radical count. The second widths satisfy $b_2, t_2 \ge 0$. Finally $A/M \le 113/125$, and direct differentiation of (7) gives

$$1 - \rho - \delta A/M \ge 259/16875, \qquad 5 - 6\rho \le 23/7.$$

For $a = M/2000$ and $\eta \le M/1000$, (9) is therefore less than $-M/100$. This pays the higher rows with ample room for the remaining small losses.

For completeness, the other source branches have independent reserves. Comparison rectangles have old capacity at most $.93M + o(M)$. For amplifier valuations $1, 6, 7$, use (5) with $r = 1/6$; the smallest error reserve is $13M/120000$. The exact first exceptional calculation gives $F_1 \ge 2c/3 - 4a - 5\varpi/6$ and the second gives $F_2 \ge 2b_2/3$. Their centered loss fits within $M - A > M/10$; zero-frequency terms and the original amplifier errors do also. No extra count of amplifier primes is introduced. These estimates, with the exact normalizers and centered cancellation, complete Proposition 1.2; see Supplement G for the full ledger.

4 Insertion into the physical numerator

The local estimate is needed only near one physical geometry. Set

$$\ell = \frac{626701}{3750000}, \qquad h = \frac{3969737987}{5000000000}, \qquad x = 1 + \ell - h, \qquad y = h - 2\ell. \tag{10}$$

Then $\theta = 11/12 - \ell/4$, $x + y = 1 - \ell$, and $0 < \ell < x < y < 2x$. The exact physical coefficient identity makes the fixed base character the source principal character outside its finite bad-prime set. Its moving part is the sextic row, so the preceding fixed-family proposition applies with the original masks.

Let d now denote a physical row exponent, and let $t = \log_U h_0$ be a free Euler shift. Near the upper row,

$$h - 10^{-3} \le d \le h + 10^{-10}, \qquad 0 \le t \le 10^{-3},$$

the upper reflected plain length and target slot capacity are

$$n(d,t) = 1 - y/d + t, \qquad z_A(t) = \frac{1 - 2n(d,t)}{C}.$$

A small capacity decrement pays scale-box errors. The exact geometric checks leave strict room inside

$$.41 < n(d,t) \le 211/500, \qquad .03 < z_A(t) \le 71/2000.$$

In an amplitude class, let $q = \delta p$ be the average slot gain, $0 \le p \le 1/2$. The source selection theorem chooses positive-gain main slots with actual width $z_I \le z_A$ such that

$$|Q_I|^2 \ge U^{2qz_A - \delta\,\mathrm{mesh} - o(1)}. \tag{11}$$

The gain is measured at the target capacity even if the selected product is shorter. The total available slot length is ℓ/d; the recorded geometry supplies the required capacity.

This distinction resolves the apparent difficulty with the lower bound $z \ge .03M$ in Proposition 1.2. If $z_I < .03$, the old moment applies, because

$$x_1 + x_2 + \tfrac{9}{2} z_I \le 2(211/500) + \tfrac{9}{2}(.03) = 979/1000 < 1.$$

If $z_I \ge .03$ but one actual plain length is below .41, the old capacity instead has reserve $33/4000$. In the remaining case both plains and the selected width lie in the local window and $x_1 + x_2 + C z_I \le 1$. The empty product uses the unrestricted zero-slot estimate. Thus every case has the same required fourth-moment bound, with the gain in (11) intact (Supplement H).

The Euler expansion must also cover every free shift. Its reflected scale and coefficient mass satisfy

$$X_u \ll_S (U/Y)h_0, \qquad \sum_{h_0,h_r} |c_u(h_0,h_r)|h_0^{\delta/2} \ll U^{\varepsilon}. \tag{12}$$

The repeated-prime factor h_r is already charged in the combined conductor; its deficit is not charged again. For $t \le 10^{-3}$, $q \le \delta/2$ gives $-\delta t/2 - qz_A(t)/2 \le -qz_A(0)/2$. For larger shifts set $b = 2y/d - 1$ and

$$z_o(t) = \max\{0, (b-2t)/(9/2)\}, \qquad S_b(t) = \frac{t}{2} - \frac{b/C - z_o(t)}{4}.$$

The exact audit gives $S_b(10^{-3}) > 3/10000$. Its slope is $7/18$ until the old capacity vanishes and $1/2$ thereafter. Hence $-\delta t/2 - qz_o(t)/2 \le -qz_A(0)/2$ for every larger shift. At zero capacity, the physical zero-slot moment has no condition on the sum of the two plain lengths; the actual lengths are each below 2, within its source domain. Therefore (12) pays every Euler term. Rows below the local interval retain the old numerator estimate.

5 The exact endpoint and continuation

Write $\kappa_* = 2\theta - 1 = 1874433/2500000$. For the detector parameter $p \in [0, 1/2]$ and zero parameter $\delta \in [0, 3/4]$, put

$$V = 5/6 - \delta, \qquad D_\kappa = 3 - p - 2p/(3\kappa), \qquad B_\kappa = 2(1-p)(1-p/(3\kappa)),$$
$$c_\kappa = B_\kappa/D_\kappa, \qquad R_\kappa = 1 - \delta + \frac{\delta c_\kappa V}{2(V + \delta c_\kappa)}.$$

The physical upper-row exponent is

$$F_{C,\kappa} = \frac{1+\ell}{2} + \frac{\delta}{2} + h\left(\frac{3R_\kappa}{4} - \frac{5}{12}\right) + \delta p\left(\ell - \frac{h - 4\ell}{2C}\right). \tag{13}$$

The rational certificate proves on the complete parameter rectangle

$$F_{C,\kappa_*} \le \theta - 10^{-7}. \tag{14}$$

To verify it, set $J = VD_{\kappa_*} + \delta B_{\kappa_*}$. The audit first proves $J \ge 1/10$. Multiplying $\theta - 10^{-7} - F_{C,\kappa_*}$ by J leaves a rational polynomial. On 18 rational subrectangles all its Bernstein coefficients are nonnegative, and their union is the whole rectangle. This proves a continuous inequality, not a numerical-grid observation. Moreover $P(\ell) > 0$ for the quartic P in Section 1, while $P(1/6) < 0$ and $P' > 0$ on $[1/6, 1/5]$. Hence $\ell > \ell_0$ and $\theta < b_0$.

The remaining physical rows require their own checks. The general exponent is affine in d for $d \ge 1/2$. At $\kappa = \kappa_*$ the certificates give the following savings below θ:

Row exponent	Physical coefficient	Saving
h	C	10^{-7}
$h - 10^{-3}$	$6\kappa_*$	10^{-5}
$1/2$	$6\kappa_*$	10^{-3}

At the junction the local exponent is no larger than the old one. Affinity covers the intervening rows, and a slope bound of 2 pays the extension to $h + 10^{-10}$. Below $1/2$, convexity and an elementary

bound at $d_0 = 1/200$ cover the retained rows. Floor classes have reserve exceeding $1/100$; very small rows have a negative exponent relative to the principal scale. The raw large-row bound pays rows above $h + 10^{-10}$ after choosing its contour parameter. The principal residue, compensated probe, and off-height tails retain the original contour construction. The exact inequalities and source domains are recorded in Supplement I.

It remains to explain why this improves the zero supremum without assuming the desired conclusion. Let β_* be the source Hecke zero supremum, including the sentinel $1/2$. The incumbent theorem gives $\beta_* \le 7/8$. Suppose $\mu = \beta_* - \theta > 0$ and choose

$$\kappa = 2\beta_* - 1, \qquad \kappa_* < \kappa \le 3/4.$$

This meets the old slot-zero hypothesis by equality. For the local branch, only the detector varies with κ; the source derivative bounds give $\partial_\kappa F_{C,\kappa} < 1/50$. Thus

$$F_{C,2\beta_*-1} - \beta_* \le -10^{-7} - \frac{24}{25}\mu.$$

On the old branches the reciprocal capacity $1/(6\kappa)$ also varies. Its separate derivative estimate is less than $1/8$, leaving at least $3\mu/4$ in addition to their certificate reserves. Every retained class consequently has a common positive saving.

Choose the geometric margins, capacity decrement, amplitude mesh, and finite slot system before fixing a primitive target η. Their smallest slot fixes the principal prime-product saving. For that target, now choose the finite dictionary and profile orders, then a sufficiently small analysis-height exponent, and finally the tail order and eventual threshold. The constants and threshold may depend on η; the resulting positive exponent Δ does not. Taking $\Delta < \mu/4$ and smaller than all the preceding reserves gives

$$J_\eta(Z) - f_\eta(Z) \ll_\eta Z^{\beta_*-k-\Delta}, \qquad k = 5/6 - x/3 - \ell/6.$$

The compensated principal bound is $J_\eta(Z) \ll_\eta Z^{\theta-k+\varepsilon}$, so the same improved exponent bounds f_η. Its Mellin transform is holomorphic in $\Re s > \beta_* - \Delta$ and, originally in $\Re s > 1$, equals

$$\int_0^\infty f_\eta(Z) Z^{-s+k} \frac{dZ}{Z} = e^{(s-5/6)^2} \frac{H_\eta(s)}{L^S(s,\eta)}.$$

The source cutoff has $|H_\eta - 1| \le 1/2$ in $\Re s > \theta$. Holomorphic continuation therefore excludes zeros to the right of $\beta_* - \Delta$. Since Δ is target-independent, choose a primitive zero sufficiently near the supremum and then use its target's estimate: this contradicts the definition of β_*. Thus $\beta_* \le \theta$.

Fixed Euler factors restore all imprimitive Hecke presentations without adding zeros in this positive half-plane. For an ordinary Dirichlet character χ, its Eisenstein norm lift satisfies

$$L_K(s, \eta_\chi) = L(s,\chi)L(s,\chi\chi_{-3}).$$

In $\theta < \Re s < 1$ the product is nonzero and both factors are finite, so $L(s,\chi) \ne 0$. Classical nonvanishing covers $\Re s \ge 1$ with the principal-pole exception. The modulus-one character gives the zeta case and completes the argument for Theorem 1.1.

6 Proof record and attribution

The full proof is retained in the separate supplement: B identifies the signed rows and masks; C fixes the character dictionary; D proves the unmarked input; E gives uniform Fourier transport; F treats

actual restricted children and radii; G assembles the local moment; H handles shortened scales and slots; I proves the endpoint and continuation. The original exact audits and the additional 20 rational endpoint checks are included in the accompanying source package [4].

The analytic argument is proposed relative to the pinned framework. Supporting development checks compile 15 Lean components and an audit module using cached dependencies, with 48 printed axiom checks containing only the standard axioms. A complete formal endpoint theorem, a clean full-dependency replay, canonical-target comparison, and independent three-kernel verification remain to be established. The finite certificates and component checks have the precise scopes recorded in the package.

OpenAI supplies the initial 7/8 theorem and analytic-arithmetic framework; Levy supplies the weighted-numerator input; Collet supplies the variable moment, detector, quartic optimization, and source architecture. The contribution proposed here is the local 4.498 refinement, its joint-label analysis, and the resulting endpoint assembly. Earlier reconstruction and unmarked-input work is recorded in [5]. The package retains Apache-2.0 licensing and upstream notices; attribution does not imply endorsement.

References

[1] Hailey Collet, *The quartic QRH boundary*, 10 October 2026. Pinned manuscript and source, commit `58344dfdbe756cf2f743da1908ffe0582418cf5b`. References B0, B3, and B4 in the supplement use this source.

[2] OpenAI, *OpenAI/math*, pinned formal development, commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`.

[3] Akash Levy, *Weighted-numerator proof*, pinned source package, commit `2fd60c0926b66ea18d7436f5ed55250fd006ab5d`.

[4] Zhongpai Gao, with AI assistance, *Local physical moment research candidate: 0.8748866*, 11 October 2026. Original research notes and certificates, commit `7f9166e2d626586cc3c89853497c0e0e22e856ee`; pull request 15. Analytic sources and component evidence, commit `44e93fbcac36bfd10cc1316674c970747888d448`.

[5] Zhongpai Gao, with AI assistance, *Joint-label reconstruction and actual unmarked inputs*, 11 October 2026. Companion source notes and supporting components.